Exercise 11

Which of the following are valid probability density functions?

1. $p_X(x) = e^{-x}1(x)$
2. $p_X(x) = e^{-|x|}$
3. $p_X(x) = \begin{cases} \frac{3}{4}(x^2-1) & |x| < 2 \\ 0 & \text{otherwise} \end{cases}$
4. $p_X(x) = 2xe^{-x^2}1(x)$

Properties:

- $p_X(x) \geq 0$ $C_a = 1$
- $F_X(x) = \int_{-\infty}^{x} p_X(z)dz$
- $\int_{-\infty}^{\infty} p_X(x)dx = 1$ $C_b = 1$
- $P(x_1 < X \leq x_2) = \int_{x_1}^{x_2} p_X(z)dz$ $C_c = 1$

1 $p(x) = e^{-x}1(x)$

$\forall_{x<0, x\in R} 1(x) = 0 \Rightarrow \forall_{x<0} e^{-x}1(x) = 0 = C_a = 1$

$\int_0^{\infty} e^{-x}1(x)\,dx = \int_0^{\infty} -e^{-x} = 0-(-1) = 1 \Rightarrow C_b = 1$

$F(x) = (1-e^{-x})1(x)$ $\int_{x_1}^{x_2}(1-e^{-x})dx = e^{-x_1} - e^{-x_2}$

$P(x_1 < X \leq x_2) = (1-e^{-x_2}) - (1-e^{-x_1}) = e^{-x_1} - e^{-x_2} \Rightarrow C_c = 1$ TAK

2. $p(x) = e^{-|x|}$ $\forall_{x\in R} e^{-|x|} > 0 \Rightarrow C_a = 1$ NIE

$\int_0^{\infty} e^{-|x|}dx = 1 + \int_{-\infty}^{0} e^{-|x|}dx = 1 \Rightarrow \int_{-\infty}^{\infty} e^{-|x|} = 2 \Rightarrow C_b = 0$

3. $p(x) = \begin{cases} \frac{3}{4}(x^2-1), & |x| < 2 \\ 0, & \text{Otherwise} \end{cases}$

$\int_0^2 \frac{3}{4}(x^2-1)\,dx = \frac{3}{4}x^3 - x \Big|_0^2 = 2-2 = 0$

NIE

$\frac{3}{4}(x^2-1) \geq 0$ $\frac{3}{4}x^2 \geq 1$

$x \in (-2,2)$ $x^2 \geq \frac{4}{3}$ $|x| \geq \sqrt{\frac{4}{3}}$ $|x| \geq \frac{2}{3}\sqrt{3}$ $x \in R - (-\frac{2}{3}\sqrt{3}, \frac{2}{3}\sqrt{3}) \Rightarrow C_a = 0$

4. $p(x) = (2xe^{-x^2})1(x)$ $\forall_{x\in R, x<0} 1(x) = 0 \wedge \forall_{x>0} 2xe^{-x^2} > 0 \Rightarrow C_a = 1$

$\int_0^{\infty} 2xe^{-x^2}dx = -e^{-x^2}\Big|_0^{\infty} = 0-(-1) = 1 \Rightarrow C_b = 1$

$F(x) = \int_{-\infty}^{x} 2xe^{-x^2}1(x)\,dx = 1 - e^{-x^2}$ $\forall_{x_2 \geq x_1}\left(F(x_2) - F(x_1) = e^{-x_1^2} - e^{-x_2^2} > 0\right)$

$C_c = 1$